

## Problem

Obtain  $f(t)$  for the following transforms:

$$(a) \quad F(s) = \frac{(s+3)e^{-6s}}{(s+1)(s+2)}$$


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$$(b) \quad F(s) = \frac{4 - e^{-2s}}{s^2 + 5s + 4}$$


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$$(c) \quad F(s) = \frac{se^{-s}}{(s+3)(s^2+4)}$$

## Step-by-step solution

## Step 1 of 16

(a)

Consider the following transform:

$$F(s) = \frac{(s+3)e^{-6s}}{(s+1)(s+2)}$$

$$\text{Let } X(s) = \frac{s+3}{(s+1)(s+2)}.$$

Hence,

$$F(s) = X(s)e^{-6s}$$

Consider the time shift property of the Laplace Transform.

$$f(t-a)u(t-a) \Leftrightarrow e^{-as}F(s)$$

Hence, the Inverse Laplace transform of  $F(s)$  is,

$$f(t) = x(t-6)u(t-6) \dots\dots (1)$$

## Step 2 of 16

Consider the partial fraction expansion for  $X(s)$ .

$$\frac{s+3}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2} \dots\dots (2)$$

Find the value of A.

$$\begin{aligned} A &= (s+1)X(s)\Big|_{s=-1} \\ &= (s+1)\frac{s+3}{(s+1)(s+2)}\Big|_{s=-1} \\ &= \frac{s+3}{(s+2)}\Big|_{s=-1} \\ &= \frac{-1+3}{(-1+2)} \end{aligned}$$

$$= 2$$

## Step 3 of 16

Find the value of  $B$ .

$$\begin{aligned}
 B &= (s+2)X(s)\Big|_{s=-2} \\
 &= (s+2)\frac{s+3}{(s+1)(s+2)}\Big|_{s=-2} \\
 &= \frac{s+3}{(s+1)}\Big|_{s=-2} \\
 &= \frac{-2+3}{(-2+1)} \\
 &= -1
 \end{aligned}$$

**Step 4** of 16

Hence, from equation (2),

$$X(s) = \frac{2}{s+1} - \frac{1}{s+2}$$

Apply Inverse Laplace transform.

$$x(t) = 2e^{-t} - e^{-2t}$$

From equation (1),

$$\begin{aligned}
 f(t) &= x(t-6)u(t-6) \\
 &= [2e^{-(t-6)} - e^{-2(t-6)}]u(t-6)
 \end{aligned}$$

Thus,  $f(t)$  of the transform is  $\boxed{[2e^{-(t-6)} - e^{-2(t-6)}]u(t-6)}$ .

**Step 5** of 16

(b)

Consider the following transform:

$$F(s) = \frac{4 - e^{-2s}}{s^2 + 5s + 4}$$

The function is written as,

$$\begin{aligned}
 F(s) &= \frac{4 - e^{-2s}}{(s+1)(s+4)} \\
 &= \frac{4}{(s+1)(s+4)} - \frac{e^{-2s}}{(s+1)(s+4)}
 \end{aligned}$$

Consider the partial fraction expansion for  $X(s)$ .

$$F(s) = \left(\frac{A}{s+1} + \frac{B}{s+4}\right) - \left(\frac{C}{s+1} + \frac{D}{s+4}\right)e^{-2s} \dots\dots (3)$$

**Step 6** of 16

Find the value of  $A$ .

$$\begin{aligned} A &= (s+1) \frac{4}{(s+1)(s+4)} \Big|_{s=-1} \\ &= \frac{4}{s+4} \Big|_{s=-1} \\ &= \frac{4}{-1+4} \\ &= \frac{4}{3} \end{aligned}$$

**Step 7** of 16

Find the value of  $B$ .

$$\begin{aligned} B &= (s+4) \frac{4}{(s+1)(s+4)} \Big|_{s=-4} \\ &= \frac{4}{s+1} \Big|_{s=-4} \\ &= \frac{4}{-4+1} \\ &= -\frac{4}{3} \end{aligned}$$

**Step 8** of 16

Find the value of  $C$ .

$$\begin{aligned} C &= (s+1) \frac{1}{(s+1)(s+4)} \Big|_{s=-1} \\ &= \frac{1}{s+4} \Big|_{s=-1} \\ &= \frac{1}{-1+4} \\ &= \frac{1}{3} \end{aligned}$$

**Step 9** of 16

Find the value of  $D$ .

$$\begin{aligned} D &= (s+4) \frac{1}{(s+1)(s+4)} \Big|_{s=-4} \\ &= \frac{1}{s+1} \Big|_{s=-4} \\ &= \frac{1}{-4+1} \\ &= -\frac{1}{3} \end{aligned}$$

**Step 10** of 16

From equation (3),

$$F(s) = \left( \frac{4}{3(s+1)} - \frac{4}{3(s+4)} \right) - \left( \frac{1}{3(s+1)} - \frac{1}{3(s+4)} \right) e^{-2s}$$

Apply Inverse Laplace transform by using time shift property.

$$\begin{aligned} f(t) &= \left[ \frac{4}{3} e^{-t} u(t) - \frac{4}{3} e^{-4t} u(t) \right] - \left[ \frac{1}{3} e^{-(t-2)} u(t-2) - \frac{1}{3} e^{-4(t-2)} u(t-2) \right] \\ &= \frac{4}{3} u(t) [e^{-t} - e^{-4t}] - \frac{1}{3} u(t-2) [e^{-(t-2)} - e^{-4(t-2)}] \end{aligned}$$

Thus,  $f(t)$  of the transform is  $\boxed{\frac{4}{3} u(t) [e^{-t} - e^{-4t}] - \frac{1}{3} u(t-2) [e^{-(t-2)} - e^{-4(t-2)}]}$ .

#### Step 11 of 16

(c)

Consider the following transform:

$$F(s) = \frac{s e^{-s}}{(s+3)(s^2+4)}$$

$$\text{Let } X(s) = \frac{s}{(s+3)(s^2+4)}.$$

Hence,

$$F(s) = X(s) e^{-s}$$

Consider the time shift property of the Laplace Transform.

$$f(t-a)u(t-a) \Leftrightarrow e^{-as} F(s)$$

Hence, the Inverse Laplace transform of  $F(s)$  is,

$$f(t) = x(t-1)u(t-1) \dots\dots (4)$$

#### Step 12 of 16

Consider the partial fraction expansion for  $X(s)$ .

$$\frac{s}{(s+3)(s^2+4)} = \frac{A}{s+3} + \frac{Bs+C}{s^2+4}$$

$$s = A(s^2+4) + (Bs+C)(s+3)$$

$$s = (As^2 + 4A) + (Bs^2 + 3Bs + sC + 3C)$$

$$s = s^2(A+B) + (3B+C)s + 3C + 4A$$

Compare the constant coefficients.

$$3C + 4A = 0$$

$$C = -\frac{4}{3}A$$

Compare the  $s^2$ -coefficients.

$$A + B = 0$$

$$B = -A$$

#### Step 13 of 16

Compare the  $s$ -coefficients.

$$3B + C = 1$$

$$-3A + C = 1$$

Substitute  $-\frac{4}{3}A$  for  $C$ .

$$-3A - \frac{4}{3}A = 1$$

$$\frac{-9A - 4A}{3} = 1$$

$$\frac{-13A}{3} = 1$$

$$A = -\frac{3}{13}$$

#### Step 14 of 16

The value of  $B$  is,

$$B = -A$$

Substitute  $-\frac{3}{13}$  for  $A$ .

$$B = -\left(-\frac{3}{13}\right)$$

$$= \frac{3}{13}$$

The value of  $C$  is,

$$C = -\frac{4}{3}A$$

Substitute  $\frac{3}{13}$  for  $A$ .

$$C = -\frac{4}{3}\left(-\frac{3}{13}\right)$$

$$= \frac{4}{13}$$

#### Step 15 of 16

Hence,

$$X(s) = \frac{-\frac{3}{13}}{(s+3)} + \frac{\frac{3}{13}s + \frac{4}{13}}{s^2 + 4}$$

$$= -\frac{3}{13(s+3)} + \frac{3s}{13(s^2+4)} + \frac{4}{13(s^2+4)}$$

$$= -\frac{3}{13(s+3)} + \frac{3}{13}\left(\frac{s}{s^2+4}\right) + \frac{2}{13}\left(\frac{2}{s^2+4}\right)$$

Apply inverse Laplace transform.

$$x(t) = -\frac{3}{13}e^{-3t}u(t) + \frac{3}{13}\cos(2t)u(t) + \frac{2}{13}\sin(2t)u(t)$$

$$= \frac{1}{13}u(t)\left[-3e^{-3t} + \frac{3}{13}\cos(2t) + \frac{2}{13}\sin(2t)\right]$$

#### Step 16 of 16

Recall equation (4).

$$\begin{aligned}f(t) &= x(t-1)u(t-1) \\&= \left\{ \frac{1}{13} u(t-1) \left[ -3e^{-3(t-1)} + 3\cos 2(t-1) + 2\sin 2(t-1) \right] \right\} u(t-1) \\&= \frac{1}{13} u(t-1) \left[ 3e^{-3(t-1)} + 3\cos 2(t-1) + 2\sin 2(t-1) \right]\end{aligned}$$

Thus,  $f(t)$  of the transform is  $\boxed{\frac{1}{13} u(t-1) \left[ -3e^{-3(t-1)} + 3\cos 2(t-1) + 2\sin 2(t-1) \right]}.$